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6th of October

ERUCA SYNC

Group 10206

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Semester 2
2025/2026

Key words : greenhouse – agriculture – automation – transmission

Abstract

Egypt faces major drawbacks that negatively affect its way to development. One of these drawbacks is climate change which refers to the vast changes in the average temperature per year. As a result, it impacts decreasing of the industrial and technological bases, leading to a deviation in Egyptian crop production. The purpose of this study is to solve the problem of the increasing threats of climate change to Siwa's crop production. To achieve that, a greenhouse project was chosen, depending on smart feedback loop system that uses temperature, soil moisture and light sensors in addition to actuators. Particularly, the structure is a gabled greenhouse where the plants must be planted and actuators are used. Moreover, the prototype had to meet some design requirements such as the parameter stabilizing time must be less than 30 min and the ability to operate within a time of at least 6 hours in addition to its ability to operate in equal time intervals for 5 consecutive days with a planting area of less than 0.25 m^2. The prototype was judged to be successful, and it met all design requirements as it took maximum 27.5 min to stabilize the parameters values, it was able to operate within 6 hours and in equal time intervals for 5 consecutive days. In addition, it had a planting area of 0.11 m^2.

Introduction

Table 1: Impact of temperature on crop yield (Moghazy & Kaluarachchi, 2021)

Crop	Change in Crop Yield (%) at Different Increases in Temperature			
	1 °C	2 °C	3 °C	4 °C
Wheat	−5.08	−9.35	−13.11	−17.65
Date Palm	−10.17	−20.38	−30.55	−40.75
Olives	−6		−14	−18
Maize		−14.4		−24.2
Barley		−17.29		−29.32

Egypt faces major challenges that hinder its development and affects its individuals in several fields, involving industry, agriculture, and even its climate. Climate change is one of them that refers to the vast changes in the average temperature per year. Hence, it affected the industrial and agricultural bases, naming it as another grand challenge. Consequently, crop production of Egypt has diminished due to heat stress and deviations in rainfall patterns that can negatively impact crop yields, particularly for heat-sensitive crops leafy greens (Abd El Maksoud, 2022). **As shown in table 1**, a small increase in temperature, like **2°C** and **4°C,** causes large hardy crops like maize losses **14.4%** and **24.2%** of their yield, respectively. Siwa Oasis is expected to lose its crop productivity due to heat changes. Using climate models, summer temperatures in Siwa are expected to rise by up to **4.65°C**. Because of this heat, plants will need **18.2%** more water just to stay alive, leading to crop failure (Moghazy & Kaluarachchi, 2021). As a result, the specific problem to be solved is the increasing threats of climate change to Siwa's crop production.

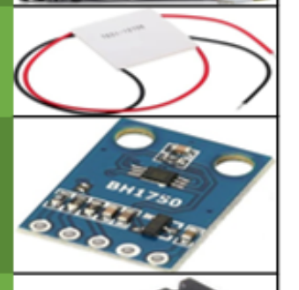
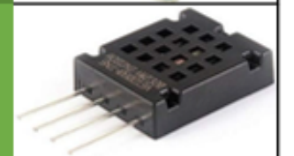
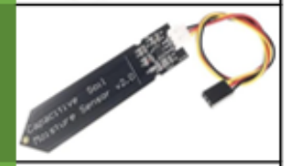
One of the prior solutions for such a problem is Thanet Earth, a massive commercial greenhouse complex conceived in the UK with a weak side of high initial investment. Nevertheless, one of its advantages is that it has vast pest and weather resistance. Another solution is El Bustan aquaponics farm, an aquaponic farm producing high quality plants, though having a weak heat control.

Hence, since the prior solutions had high initial cost and weak control, the selected solution is a low-cost gabled greenhouse with light, moisture and temperature control. Particularly, the prototype consists of a wooden gabled greenhouse structure with a maximum height of 0.45m and 0.55m, 0.45m for width and length respectively, to hold plant growth. Thereby, the structure is implemented to a smart closed loop system that associates all electric components with the ESP32 Microcontroller. Furthermore, temperature, soil moisture, light sensors and actuators were employed within the prototype to maintain the environmental conditions for plants to grow. Therefore, the project must meet some requirements to emphasize its successful working. They include its maximum time to stabilize a parameter reading that must not exceed 30 minutes. In addition, it must be able to operate within a time of at least 6 hours, as well as its ability to operate at equal time intervals for 5 consecutive days. Furthermore, the planting area must not exceed 0.25 m2. Finally, it is required to maintain the databases for two plants in the project. Several materials with different properties were used to meet the past requirements and achieve the success of the project, as shown in the next section.

Materials & Methods

Materials

Table 2, Materials of the prototype

Name	Description	Quantity	Picture
Old Recycled Wood	Rigid old recycled wood pieces.	4 Old Furniture Pieces	
Recycled Plastic Sheets	Old recycled plastic sheets used before to package things.	1.5 m^2	
MDF spray	A strong adhesive spray provides fast drying and an even adhesive layer.	1 Unit	
Servo Motors	Geared motor with position feedback	3 Units	
Computer Fans	Motorized blades in a protective frame	2 Units	
Thermoelectric Cooler (Peltier)	Ceramic-based solid-state heat pump	1 Unit	
BH1750 Digital Light Intensity Sensor	Digital light-measuring integrated circuit	1 Unit	
AM-2320 Temperature Sensor	Digital temperature module	1 Unit	
Capacitive Analog Soil Moisture Sensor	Corrosion-resistant moisture-measuring probe	1 Unit	
ESP 32-S3	Wi-Fi/Bluetooth dual-core microcontroller	1 Unit	
Mosfet Kit	Transistors with associated cooling hardware	3 Units	
Power Supply	AC-to-DC voltage converter	1 Unit	
NeoPixel Stick	Linear array of addressable LEDs	6 Units	

Methods

1-The greenhouse is constructed using recycled components for the main structure. **As shown in Table 2**, the materials are divided into structural components and the electronic feedback system components.
2-Before constructing, a 3D CAD model of the structure **(Figure 1)**, and a circuit schematic was designed for visualization and mapping the connections.

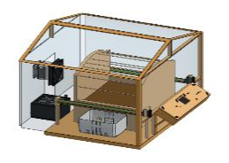


Figure 1: 3D CAD model for the prototype

Sensor	Raw Sensor Output	Reference Value
Soil Moisture	550 ± 1	0%
	194 ± 1	100%
Light Intensity	0 lx ± 1 lx	0 lx ± 1 lx
	20.03 lx ± 1 lx	20 lx ± 1 lx
Temperature	10.8 °C ± 0.5 °C	11 °C ± 0.5 °C
	21.2 °C ± 0.5 °C	20.5 °C ± 0.5 °C
	38.8 °C ± 0.5 °C	39 °C ± 0.5 °C

Table 3: Sensor Calibration Summary and Error

3-The soil moisture, light intensity, and temperature sensors were calibrated against standard reference values, with the calibration process recorded on video. The raw sensor outputs, corresponding reference values, and the quantified error margins are summarized in **Table 3**.
4- To ensure our greenhouse feedback loop operates with maximum accuracy, all three environmental sensors were calibrated. A simple two-point test was used for the soil moisture sensor **(Figure 2),** measuring it in dry air for a 0% baseline and then in a full cup of water for a 100% maximum. Defined by the equation
5-The structural frame **(Figure 5),** including the base, vertical columns, and two roof arcs connected by crossbeams, was assembled using MDF, nails, and recycled furniture wood. A plastic-lined wooden drawer with drilled drainage holes **(Figure 6)** was centered on the base as the planting bed, and filled with a two-thirds sand and one-third peat moss growing medium **(Figure 7).**
6-Wooden columns were mounted to hold four servo motors **(Figure 8),** which control PVC pipes wrapped in variable-transmissivity plastic for sunlight regulation. For thermal control, a Peltier plate with a heat sink and fan was housed within the side panels. Finally, a wooden control panel was built for the main circuit **(Figure 9),** and the water pump tubing was routed from an external tank to the soil.

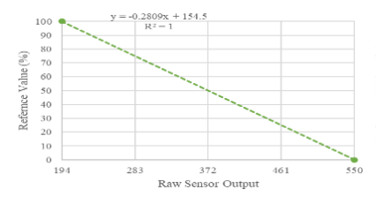


Figure 2: Calibration Curve of Soil Moisture Sensor

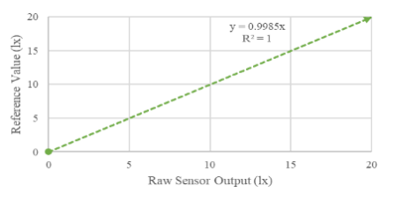


Figure 3: Calibration Curve of Light Intensity Sensor

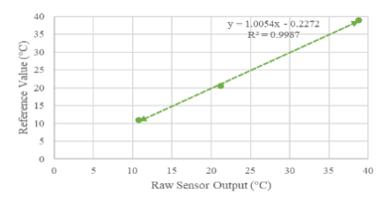


Figure 4: Calibration Curve of the Temperature Sensor

Figure 5: Constructing the outer frame

Figure 7: Filling the planting area with soil mixture

Figure 6: Making the drainage holes

Figure 8: Attaching the servo motors holders

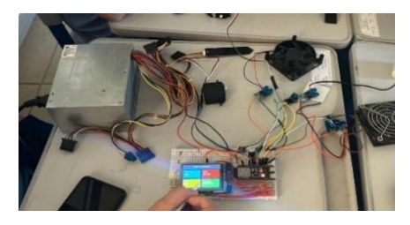


Figure 9: Feedback system's circuit

Test Plan

1-The planting area should be measured to make sure it doesn't exceed the required 0.25m^2. Furthermore, the greenhouse prototype ran automatically for a full 6 hours while the live sensor readings were being recorded. During this phase, the specific actuators, the Peltier fans, the water pump, and the sunlight controlling system, were artificially triggered to verify their responsiveness, and the power consumption of each actuator was measured and reported in watts.
2-The system's feedback control loop was evaluated by introducing three realistic deviations from normal conditions for the selected three parameters: Temperature, light, and moisture. Moreover, the time required for the control algorithm to restore system stability and its maximum was recorded. In addition, while the prototype was placed outdoors, the data logging was conducted four times daily at fixed intervals for five consecutive days and then the average of three readings were recorded.
3- A preloaded database for containing environmental condition ranges for arugula and spinach was set up in the software of the prototype. The testing involved switching the active profile to verify that there is no deficiency in the data or updates.

Results

Negative results:

1) Two servo motors were permanently damaged due to a wiring error. Accidentally swapping the Ground (GND) and Power (VCC) jumper wires led to an immediate short circuit and thermal overload that destroyed the motors.
2) The horizontal crossbeams connecting the upper roof arcs snapped completely during assembly. They were cut too thin, which lacked the necessary strength to withstand the weight applied when mounting the PVC curtain mechanism.
3) The soil moisture sensor returned negative percentage readings when exposed to ambient dry air, despite being calibrated to a 0% baseline. To resolve this hardware issue, a software adjustment was integrated into the microcontroller's logic, ensuring that any calculated value dropping below zero was automatically constrained to exactly 0%.

Positive results:

1-**Table 4** confirmed the feedback loop's efficiency, successfully restoring system stability with maximum recovery times of 1,650 seconds for temperature shifts, 45 seconds for light changes, and 12 seconds for soil moisture deviations.
2-**Figure 10** shows that the system successfully maintained stability across all three parameters during continuous operation.
3- **As shown in table 5**, the power total power consumption of the actuators was around 41 w.
4) **As shown in Figures 11, 12, and 13,** and **table 6**, the prototype's consistency over five consecutive days. Accurately tracking distinct cycles at 6:00 AM, 2:00 PM, and 10:00 PM.

Table 4: Time to restore system stability

Parameter	Deviation	Time to restore system stability	Additional alerts or actions
Soil Moisture	Soil Moisture < 5%	1 sec.	Pumps turn off and alert is sent: "Soil moisture sensor is probably disconnected"
	Soil Moisture = 49%	12 sec.	Pumps turn on for a specific time.
	Soil Moisture > 85%	1 sec.	Pumps turn off and an alert is sent
Temperature	Temperature = 34.1 °C	1650 sec.	cooling system works, and fans work at 100%
	Temperature = 10 °C	1500 sec.	heating system works, and fans work at 100%
	Temperature value is not a number due to physical dysconnectivity	1 sec.	Cooling/Heating system turns off, fans turn off and an alert is sent
Light Intensity	Sunlight Intensity < 9,720 lx	1 sec.	Convert to the lower level of transmissivity or opens the LEDs
	Sunlight Intensity = 81,000 lx	45 sec.	Convert to the lower level of transmissivity
	Light intensity value is not a number due to physical dysconnectivity	1 sec.	LEDs turn off and an alert is sent.

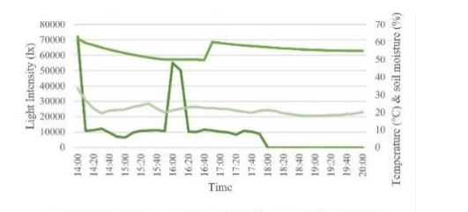


Figure 10: Temperature, Soil Moisture, and Light Intensity readings over 6 hours

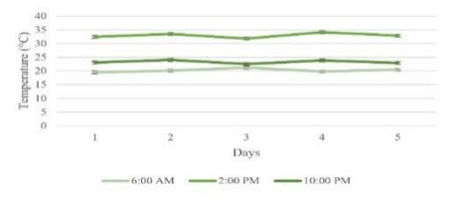


Figure 13: Average Temperature over equal time intervals on five consecutive days

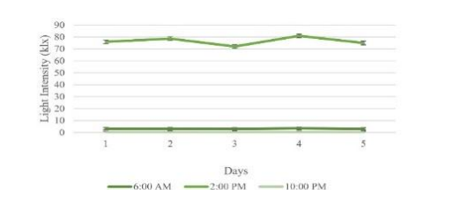


Figure 12: Average Light Intensity over equal time intervals on five consecutive days

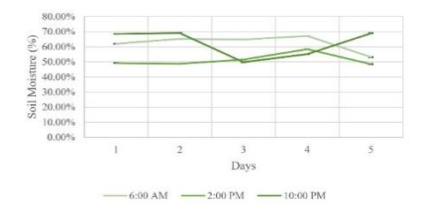


Figure 11: Average Soil Moisture over equal time intervals on five consecutive days

Table 5: Time to restore system stability

Actuator	Voltage	Current	Power Consumption
Water Pump	5 V ± 0.026 V	0.06 A ± 0.0017 A	0.3 W ± 3.35% W
2 Computer Fans	24 V ± 0.13 V	0.44 A ± 0.0093 A	10.56 W ± 2.66% W
Peltier	12 V ± 0.07V	2 A ± 0.09 A	24 W ± 5.1% W
2 Servo Motors	10 V ± 0.06 V	0.6 A ± 0.017 A	6 W ± 3.43 W
LEDs	5 V ± 0.026 V	0.08 A ± 0.0021 A	0.4 W ± 3.15% W
Total Power	41.26 W ± 1.27 W		

Table 6, response time across 5 days

Parameter / Response Time	Day 1			Day 2			Day 3			Day 4			Day 5		
	6:00 AM	2:00 PM	10:00 PM	6:00 AM	2:00 PM	10:00 PM	6:00 AM	2:00 PM	10:00 PM	6:00 AM	2:00 PM	10:00 PM	6:00 AM	2:00 PM	10:00 PM
Temperature	19.5 °C ± 0.5 °C	32.4 °C ± 0.5 °C	23.1 °C ± 0.5 °C	20.1 °C ± 0.5 °C	33.5 °C ± 0.5 °C	24.0 °C ± 0.5 °C	21.2 °C ± 0.5 °C	31.8 °C ± 0.5 °C	22.5 °C ± 0.5 °C	19.8 °C ± 0.5 °C	34.1 °C ± 0.5 °C	23.8 °C ± 0.5 °C	20.5 °C ± 0.5 °C	32.9 °C ± 0.5 °C	22.9 °C ± 0.5 °C
Response Time	0 sec.	840 sec.	0 sec.	0 sec.	867 sec.	0 sec.	0 sec.	820 sec.	0 sec.	0 sec.	882 sec.	0 sec.	0 sec.	848 sec.	0 sec.
Soil Moisture	62.1% ± 0.28%	49.2% ± 0.28%	68.5% ± 0.28%	65.2% ± 0.28%	48.7% ± 0.28%	69.1% ± 0.28%	64.8% ± 0.28%	51.55 % ± 0.28%	49.8% ± 0.28%	67.2% ± 0.28%	58.4% ± 0.28%	55.1% ± 0.28%	53.2% ± 0.28%	48.5% ± 0.28%	68.9% ± 0.28%
Response Time	0 sec.	12 sec.	0 sec.	0 sec.	13 sec.	0 sec.	0 sec.	0 sec.	12 sec.	0 sec.	0 sec.	0 sec.	0 sec.	14 sec.	0 sec.
Light Intensity	3,200 lx ± 1 lx	76,000 lx ± 1 lx	0 lx ± 1 lx	3,150 lx ± 1 lx	78,500 lx ± 1 lx	0 lx ± 1 lx	2,900 lx ± 1 lx	72,000 lx ± 1 lx	0 lx ± 1 lx	3,400 lx ± 1 lx	81,000 lx ± 1 lx	0 lx ± 1 lx	3,050 lx ± 1 lx	75,000 lx ± 1 lx	0 lx ± 1 lx
Response Time	1 sec.	44 sec.	0 sec.	1 sec.	45 sec.	0 sec.	1 sec.	33 sec.	0 sec.	1 sec.	45 sec.	0 sec.	1 sec.	34 sec.	0 sec.

Analysis

Feedback and automation system:

To achieve the best results, as shown in the results section, a feedback and automation system had to be made. **As shown in Figure 14**, the brain of the system that handles all the logic, sensor data processing, and control signals for the actuators is an ESP32-S3. It is connected to multiple sensors, which are a BH1750 digital light intensity sensor, an AM-2320 temperature sensor, and a capacitive analog soil moisture sensor V2.0, that sends environmental data via I2C or analog signals. Figure 15 shows the code that processes the data by converting the raw sensor outputs to the calibrated values according to the calibration equations using the ESP. After processing the data, the ESP sends orders to multiple connected actuators: 2 fans, 2 servo motors, 1 thermoelectric cooler, 1 water pump, and 6 LED sticks, to stabilize the system's environmental parameters according to specific conditions and ranges. To manage the power getting to the actuators from the power supply, several MOSFETs were used, which can control the flow of electricity through the actuators. In addition, the microcontroller is connected to a TFT touch screen that represents the live sensor readings and their graphs, and it can be used to switch between the environmental condition ranges according to the selected plant species.

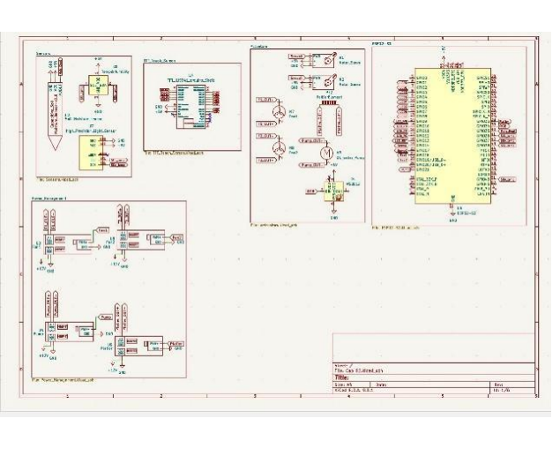
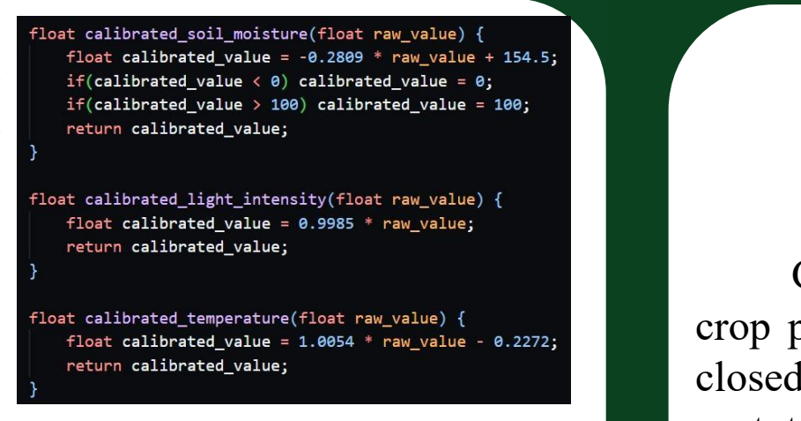


Figure 14: Circuit's schematic diagram



```
float calibrated_soil_moisture(float raw_value) {
    float calibrated_value = -0.2809 * raw_value + 154.5;
    if(calibrated_value < 0) calibrated_value = 0;
    if(calibrated_value > 100) calibrated_value = 100;
    return calibrated_value;
}

float calibrated_light_intensity(float raw_value) {
    float calibrated_value = 0.9985 * raw_value;
    return calibrated_value;
}

float calibrated_temperature(float raw_value) {
    float calibrated_value = 1.0054 * raw_value - 0.2272;
    return calibrated_value;
}
```

Figure 15: Mathematical calibration code

Expected Optimal Time of Irrigation:

It is essential to keep the soil moisture within a suitable range for the growth of he arugula and spinach plants. For the arugula plant, soil moisture of approximately 58% to 62% is considered a suitable range (Lee et al., 2024). For the spinach plant, soil moisture of approximately 65% to 75% is considered a suitable range (Sun et al., 2023). To ensure that the system takes the least amount of time to irrigate the plants and reach the optimal soil moisture percentage, the expected optimal time of irrigation should be calculated. This will help prevent unexpected errors, such as over-irrigation, and check whether the irrigation time meets the design requirements.

$$V_w = \Delta FC\% \times FC\%_{total} \times V_s$$

Equation 1: Required Volume of Irrigation Water

First, to get the time of irrigation, the required volume of irrigation water should be calculated. **As shown in Equation 1**, V_w is the volume of irrigation water, $\Delta FC\%$ is the difference between the current percentage of water in the soil's field capacity and the target percentage, $FC\%_{total}$ is the standard total field capacity for sandy loam soil, which means the percentage of empty spaces in the soil that can hold water, which is equal to 20% in the case of sandy loam (***Cornell University, 2010)***, and V_s is the volume of the soil. In the worst case of irrigating the plants, the current percentage of water in the field capacity would be 50%. So, to make it reach 60% for arugula and 70% for spinach, the difference in percentages is 10% and 20%. By substituting in Equation X, the volumes of water needed to irrigate the arugula and spinach plants are approximately 151.22 ml and 302.44 ml, respectively.

$$t = \frac{V_w}{Q}$$

Equation 2: Time of Irrigation

Equation 2 is used to calculate the time of irrigation (t) using the required water volume (V_w) and the volume flow rate of the water pump (Q), which is equal to 1333 ml/min. In the worst case. By substitution, the expected time required for the arugula and spinach irrigation is 6.8 sec. and 13.61 sec. respectively. An additional 5 sec can be added due to the time of water flow through the hose to reach the openings.

Light intensity calculations & actuating:

With its various units, light intensity parameter is a main factor that affects plant growth because of its major role in photosynthesis process. The units of light intensity are divided into 3 main types, photometric, radiometric, and quantum units. Firstly, the measurement unit of the used sensor, photometric unit, is lux unit which is equivalent to one lumen per square meter. Another type is radiometric unit, energy per unit time and unit area (W/m^2). Quantum unit can be measured with µmol . s^{-1} . m^{-2}, which is based on the number of photons in a certain wave band incident per unit time (s) on a unit area (m2) divided by the Avogadro constant. The needed light intensity for arugula plant to grow is 185:200 µmol . s^{-1} . m^{-2}. Then, as the sensor uses lux unit, the value must be converted according to **equation 3** by multiplying by the conversion factor of sun light, which is 54 ***(Thimijan & Heins, 1983)***, where L is light intensity measured with photometric and quantum units. As a result, 9990: 11610 lx is the resulting light intensity needed.

$$L_{\text{Photometric}} = F_{sunlight} \cdot L_{\text{Quantum}}$$

Equation 3, light value with lx. *(Thimijan & Heins, 1983)*

After finding the light intensity needed with photometric unit, the actuation of light was done by using plastic films with different transmissivities. Transmissivity is the ratio between the light intensity entering the greenhouse and the reached sunlight intensity **as shown in equation 4**.

$$Transmissivity = \frac{L_{entered}}{L_{sun}}$$

Equation 4, transmissivity

Then, plastic films and papers with different shades were exposed t lamp light and the light intensity was measured to get transmissivity value **shown in figure 16**. Hence, the plastic films can minimize the sunligh intensity throughout the day to reach the needed value, knowing that blac films are used to block most of the light when the light intensity ove exceeds the needed value (cannot be minimized to reach it), and that is whe the LED is turned on. Therefore, the light intensity reaches the plant is at th optimum value and provides virtuous conditions for crop growth in addition to preserved energy consumption

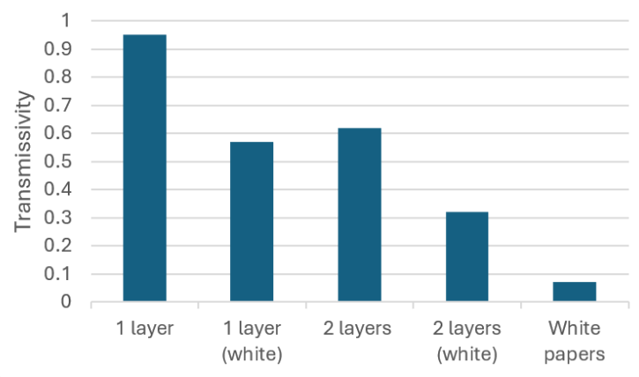


Figure 16, transmissivity for several plastic films and papers

Geometric shape analysis:

To preserve the consumption of energy in the project, geometric shape plays a major role in minimizing the range of light intensity needed to actuate and then the energy consumption of motors and lamps. Thus, the average solar radiation in the region of Siwa is for one day is 6.3 kWh/m^2/ day (Atia et al., 2023). Hence, dividing by 24 hours, the average sunlight irradiance is 0.2625 Kw/ m^2. Since the light intensity needed for the arugula plant is measured by µmol . s^{-1} . m^{-2}, the unit must be converted according to equation 5 by multiplying with the conversion coefficient of daylight (where L$_{\text{Quantum}}$ is the light intensity measured by µmol . s^{-1} . m^{-2}, and L$_{\text{Radiometric}}$ is the light intensity measured in W/m^2).

$$L_{\text{Quantum}} = C_{Daylight} \cdot L_{\text{Radiometric}}$$

Equation 5, Light value with µmol . s^{-1} . m^{-2}

After finding the suitable shape, the next step was finding the average light intensity entering the greenhouse during the year. **As shown in figure 17**, the transmissivity of the greenhouse with the selected structure averages between 0.35 (winter through north-south direction) and 0.57 (summer through north-south direction). Therefore, substituting in equation 4 (the previous analysis point), the resulted average light intensity entering the greenhouse ranges between 420 and 660 µmol . s^{-1} . m^{-2}.

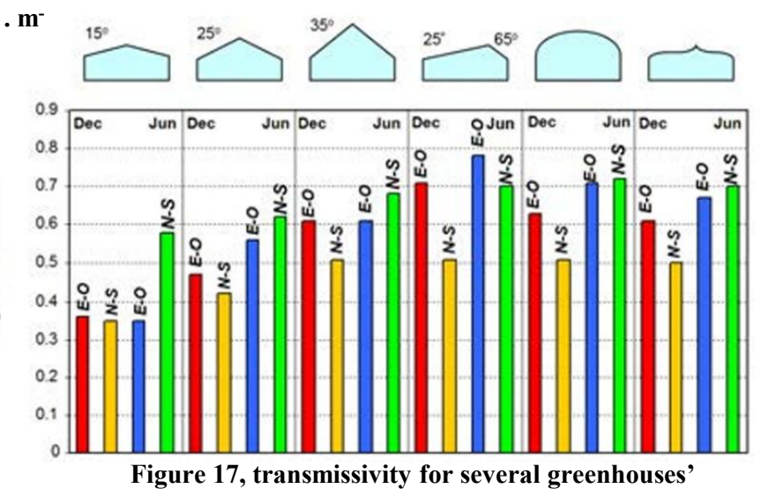


Figure 17, transmissivity for several greenhouses' geometric shape (Valera Martínez et al., 2016).

Table 7: Learning outcomes used

Subject	LO	Concepts	Usage
Physics	LO.2.10	D.C Motor	Using a DC servo motor as an actuator to control light intensity of the sun.
	LO.2.11	Step-up/step-down transformer	Utilizing transformers to output the required voltage for the electric components in the circuit.
Chemistry	LO.2.11	Greenhouse effect	Utilizing the greenhouse effect in providing the suitable environment in our system.
Mechanics	LO.2.05	Power	Calculating the total power and energy consumption of the prototype.
Biology	LO.2.09	Hormonal Controls of cycles	Simulating control systems like that in our bodies to control the environmental conditions in the system.

Conclusions

Climate change poses a severe threat to Egypt's agricultural development, specifically causing deviations in crop production in regions like Siwa. Eruca sync successfully addressed this challenge, proving that a smart, closed-loop feedback system can actively protect crops from environmental shifts. Results demonstrated that the prototype met all specific design requirements. The system proved its reliability by operating automatically for a full 6 hours and successfully logging data at equal time intervals over 5 consecutive days outdoors. Furthermore, when parameters changed, the temperature, soil moisture, and light sensors efficiently triggered the actuators to restore optimal conditions. System stability was fully achieved in just 27.5 minutes maximum, beating the required 30-minute maximum. Compared to traditional farming, which leaves crops entirely exposed to climate variations, Eruca Sync significantly minimizes crop loss. By continuously adapting to the environment without human intervention, this prototype offers to increase Egypt's agricultural base.

Recommendation

Real life application:

It is recommended to apply this project to the Siwa region, which is a natural depression with an area of 0.28 million acres located in the northwest of the Western Desert in Egypt, **as shown in Figure 18**. Egypt's proposed development project in Siwa is to reclaim 30,000 acres. However, Siwa's crop production is threatened due to global climate change. A study investigated whether government goals are possible under climate change during this century using different climate models under two representative concentration pathways (RCPs): RCP 4.5 and RCP 8.5. Using these models, summer temperatures in Siwa are expected to rise by up to 4.65 °C, which means that plants will need 18.2% more water just to stay alive. Hardy crops like maize losses 14.4% and 24.2% of their yield due to a small increase in temperature, such as 2 °C and 4 °C ***(Moghazy & Kaluarachchi, 2021)***. This means that tiny leafy greens, like arugula and spinach, would likely face 100% crop failure if planted in the same conditions. That's why this location is perfect for confronting the increasing threats of climate change to Siwa's crop production.

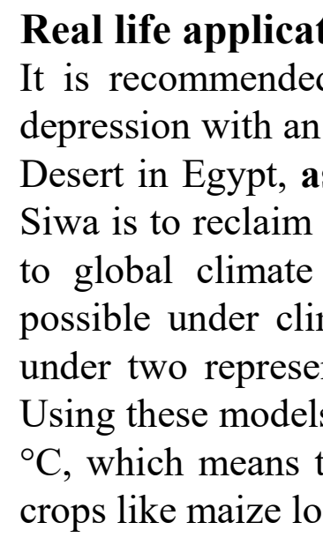
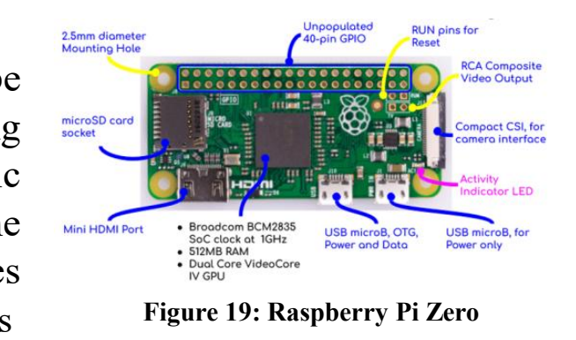


Figure 18: Reclamation Areas in the Siwa Oasis Region.

For the materials of the real-life project, the real-life project should mostly be composed of the same materials as the prototype, except for the plastic sheets covering the greenhouse, which could be replaced with isolating glass pieces. The electronic components don't have to be recycled. The ratio between the dimensions of the prototype to that of the real-life project will be 2.00387:100 to make 1,000 greenhouses across the reclamation area of the Siwa region. The real-life project's dimensions will be the same as the prototype, but with consideration of the scale conversion. The real-life project should have the ability to operate for multiple days without burning out.

Raspberry Pi Zero SBC:

For the materials of the real-life project, the real-life project should mostly be composed of the same materials as the prototype, except for the plastic sheets covering the greenhouse, which could be replaced with isolating glass pieces. The electronic components don't have to be recycled. The ratio between the dimensions of the prototype to that of the real-life project will be 2.00387:100 to make 1,000 greenhouses across the reclamation area of the Siwa region. The real-real-life project's dimensions will be the same as the prototype, but with consideration of the scale conversion. The real-life project should have the ability to operate for multiple days without burning out.



2.5mm diameter Mounting Hole
Unpopulated 40-pin GPIO
RUN pins for Reset
RCA Composite Video Output
microSD card socket
Compact CSI, for camera interface
Activity Indicator LED
Mini HDMI Port
Broadcom BCM2835 SoC clock at 1GHz
512MB RAM
Dual Core VideoCore IV GPU
USB microB, OTG, Power and Data
USB microB, for Power only

Figure 19: Raspberry Pi Zero

Shift Register Integration for Full Automation:

To improve the prototype's automation and prepare it for larger farms, expanding the system's hardware capacity is a necessity. Currently, the greenhouse relies on a user manually switching the system between day and night profiles. This manual step is necessary because the ESP microcontroller does not have enough pins to run all the physical actuators while keeping the Wi-Fi connection active. To solve this problem, we highly recommend integrating a shift register into the main circuit. A shift register is a highly effective component that uses only a few pins from the microcontroller to create multiple new output pins. By adding this chip, the water pump, Peltier module, and all servo motors can be easily controlled while keeping the necessary pins free for the Wi-Fi module.

Once the hardware is upgraded with a shift register and Wi-Fi is safely connected, the system can be fully automated. The greenhouse will be able to connect to the internet and automatically sync its day and night cycles with the actual sunrise and sunset times. Finally, this internet connection will allow developers to build a smart web dashboard, letting farmers remotely monitor temperature, moisture, and light levels from anywhere, making this prototype a scalable solution for modern agriculture.

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Acknowledgment

Thanks and gratitude go to the capstone general, Mohamed Bekheet, the capstone leader, Ninette Refaat, and the capstone teacher, Mona, for their huge contributions to the production process of this work. Without their efforts, this project wouldn't have been presented and produced in this way that satisfies the Capstone requirements and constraints.